Practice workbook · Compare light and electron microscopes and use their parts

Why magnification or an electron microscope helps, which organelles only an EM reveals, and functions of light microscope parts.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-COMPARE-LIGHT-ELECTRON-MICROSCOPE

1. Question 3 · part c · 2 marks

June 2023 Paper 2R

Worked-example retry · Attempt part c only.

3 (a) Describe what is meant by a specialised cell.

(2)

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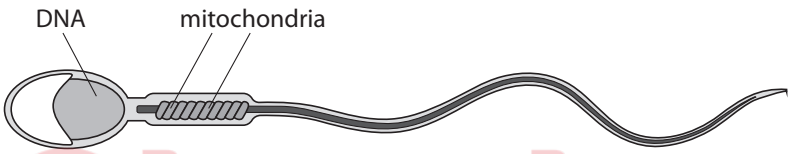
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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

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(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

(2)

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

(1)

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1. Question 3 · part c · 2 marks

June 2023 Paper 2R

Worked-example retry · Attempt part c only.

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(iii) Explain why the sperm cell has many mitochondria.

(2)

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(iv) A sperm cell travels at a speed of 0.070 mm/s.

It travels a distance of 180 mm to fertilise an egg.

Use this equation to calculate the time it takes for the sperm to reach the egg.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Give your answer in minutes.

(3)

time taken = ... minutes

(c) An electron microscope can be used to view structures in specialised cells.

Explain why an electron microscope is more useful than a light microscope to view cell structure.

(2)

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(Total for Question 3 = 12 marks)

1. Question 3 · part c · 2 marks · Working and answer

Working and answer

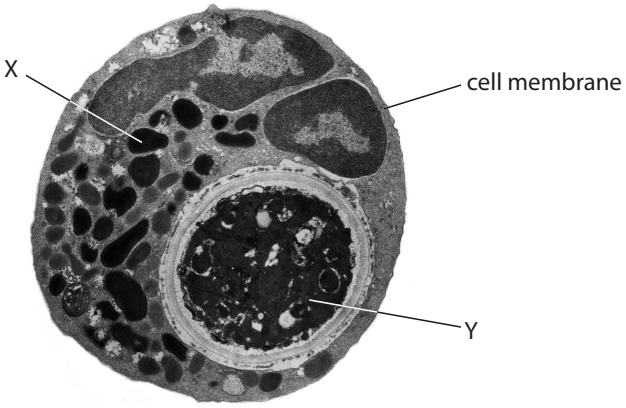
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2. Question 2 · part a.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.ii only.

2 (a) The image is an electron micrograph of a human body cell.



(Source: © Science History Images/Alamy)

(i) The table gives information about parts of the human body cell.

Complete the table by giving the missing information.

(2)

Part	Name of part	Function
X		release energy from glucose
Y	nucleus	

(ii) Electron microscopes and light microscopes can both be used to view body cells.

Which of these is an advantage of using an electron microscope?

(1)

- ☒ **A** more cell structures can be seen with greater resolution
- ☒ **B** more cell structures can be seen with less resolution
- ☒ **C** fewer cell structures can be seen with greater resolution
- ☒ **D** fewer cell structures can be seen with less resolution

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P 5 8 5 6 6 A 0 4 2 8

2. Question 2 · part a.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.ii only.

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(iii) The actual diameter of this human body cell before magnification is 0.05 mm.

Use information from the image to calculate the magnification of the cell.

(3)

magnification = ...

(b) Body cells produce carbon dioxide.

Describe how carbon dioxide passes from a body cell into the blood.

(2)

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(Total for Question 2 = 8 marks)

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Working and answer

3. Question 3 · part a.ii · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

3 (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

(2)

	Steps
	select a suitable objective lens
5	adjust focusing wheel to obtain a clear image
1	remove cells from the inside of the cheek
	place microscope slide onto the microscope stage
	smear the cell sample onto a microscope slide

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

(1)

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(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

(1)

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(iv) Give two ways that the student could avoid health hazards when working with living cells.

(2)

1 .....

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2 .....

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6

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3. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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4. Question 5 · part a.ii · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

5 The diagram shows a sperm.



(a) (i) Describe the function of a sperm.

(2)

(ii) State two other structures, apart from mitochondria, that can be seen in the sperm if it is viewed using an electron microscope.

(2)

1 .....

2 .....

(b) Explain why there are a large number of mitochondria found in a sperm.

(3)

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4. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

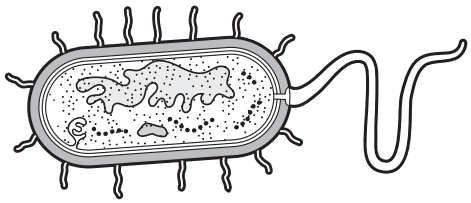
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5. Question 1 · part b.i · 3 marks

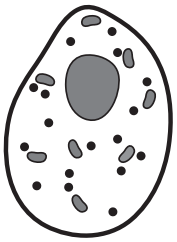
June 2024 Paper 1R

Further practice · Attempt part b.i only.

1 (a) The diagrams show a bacterial cell and a human skin cell.



Bacterial cell



Human skin cell

(i) Give **two** similarities between the bacterial cell and the human skin cell.

(2)

1 .....

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2 .....

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(ii) Give **two** differences between the bacterial cell and the human skin cell.

(2)

1 .....

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2 .....

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5. Question 1 · part b.i · 3 marks

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Further practice · Attempt part b.i only.

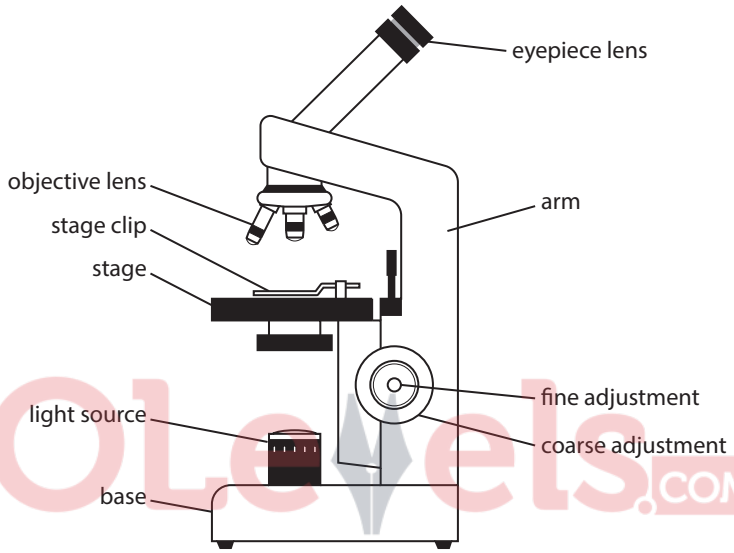
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(b) Many structures in cells can be viewed more clearly using a light microscope.

The diagram shows a light microscope.



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(i) Draw **one** straight line from each part of the light microscope to its function.

(3)

Part of microscope	Function
objective lens	holds the microscope slide in place
stage clip	moves the stage to produce a clear, focused image
fine adjustment	magnifies the sample
	keeps the microscope steady

5. Question 1 · part b.i · 3 marks

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Further practice · Attempt part b.i only.

(ii) The image size of a bacterial cell viewed using an electron microscope is 5.0 cm.

The bacterial cell has been magnified 50 000 times.

Calculate the actual size of this bacterial cell in mm.

image size = actual size × magnification

(3)

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actual size of bacterial cell = ... mm

(Total for Question 1 = 10 marks)

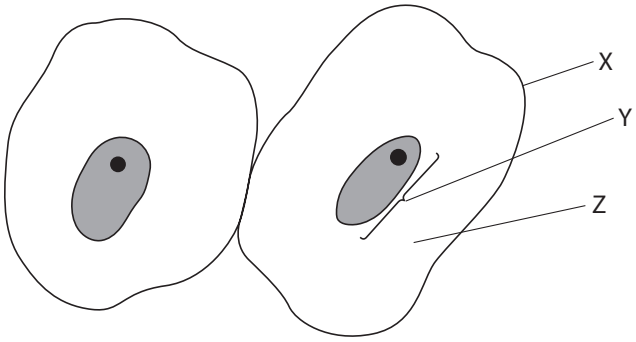
Working and answer

6. Question 7 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

7 The diagram shows two cheek cells from the inside of the mouth.



(a) Complete the table by giving the missing information.

(6)

Part	Name of part	Function of part
X		
Y		
Z		

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(b) Describe how a student would obtain and prepare these cells so the student can view them using a light microscope.

(4)

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6. Question 7 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

(c) Name two other structures inside the cells that could be seen if an electron microscope is used.

(2)

1 .....

2 .....

(Total for Question 7 = 12 marks)

Working and answer

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